Questions about regional analysis

1. ETDRS Grid area pixels calculation

In the following Matlab code, `sum(sum(sector_masks(:,:,2)))` is repeated twice at the end when creating the regional pixel vector.

```
imgArea_pix_vec = [sum(sum(foveal_mask)) sum(sum(parafoveal_mask))
sum(sum(sector_masks(:,:,1)+sector_masks(:,:,2)+sector_masks(:,:,3)))...

sum(sum(sector_masks(:,:,4)+sector_masks(:,:,5)+sector_masks(:,:,6)))
sum(sum(sector_masks(:,:,1)+sector_masks(:,:,6)))
sum(sum(sector_masks(:,:,3)+sector_masks(:,:,4)))...
                    sum(sum(sector_masks(:,:,2)))
sum(sum(sector_masks(:,:,2)))];
```

Is the last element supposed to be `sum(sum(sector_masks(:,:,5)))` based on the binary region division?

2. Double "pix_to_mm" multiplication when computing VLD_pct

`pix_to_mm` was multiplied when computing VLD_pct, however, it was multiplied when calculating totalLength too. VLD_pct is equivalent to `round(sum([app.Glinks.pixLength]*pix_to_mm)*pix_to_mm/imgArea_mm,4)*100`

```
regionalmetrics(i).totalLength =
round(sum([app.Glinks.pixLength]*pix_to_mm),2); %In mm
regionalmetrics(i).VLD_pct =
round(regionalmetrics(i).totalLength*pix_to_mm/imgArea_mm,4)*100; %in
percent
regionalmetrics(i).VLD_mminv =
round(regionalmetrics(i).totalLength/imgArea_mm,4); %in inverse mm
```

What should the formula of calculating VLD_pct be?

3. Cropping regions

In Matlab ETDRS and Foveal-ring grid types, we crop some regions.

1. What regions should be cropped? Eg. in ETDRS:

- we crop binaryReg[0] and imgReg[0] (foveal circle),
- but we don't crop binaryReg[1] (parafoveal ring consideration)

2. When we crop a region, should we crop the mask used for the area too (should we update the imgArea_pix_vec value for that region, eg. np.sum(foveal_mask))?